High Availability

Reference Patterns & Standards

Mandatory Compliance Requirements for All Systems

Document ID:	STD-HA-2026-001
Version:	2.1
Date:	May 5, 2026
Classification:	Internal - Controlled Distribution
Owner:	Enterprise Architecture - Infrastructure Standards
Status:	Approved - Mandatory Compliance
Review Cycle:	Annual (Next Review: May 2027)
Governing Body:	Architecture Review Board (ARB)

Version History

Version	Date	Author(s)	Description of Changes
2.1	05/05/2026	J. Morrison, A. Kapoor	Added circuit breaker pattern, updated multi-region requirements
2.0	01/15/2026	A. Kapoor, R. Torres	Major revision: added cloud-native patterns, AI validation criteria
1.5	09/01/2025	J. Morrison	Updated backup standards, added queue-based resilience pattern
1.4	06/12/2025	R. Torres, M. Singh	Added network resilience section, cross-DC requirements
1.3	03/20/2025	J. Morrison	Updated tiering criteria based on OCC guidance
1.2	11/05/2024	A. Kapoor	Added monitoring and alerting standards
1.1	07/18/2024	R. Torres	Added database HA standards
1.0	03/01/2024	J. Morrison	Initial document creation

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1 Purpose & Scope

1.1 Document Purpose

This document defines the mandatory High Availability (HA) patterns, standards, and requirements that all production infrastructure designs within Example Organization MUST comply with. It serves as the authoritative reference against which all Single Design Documents (SDDs) are validated prior to Architecture Review Board approval and deployment to production environments.

The standards herein are prescriptive and non-negotiable for the applicable system tiers. They represent the minimum acceptable HA posture derived from regulatory requirements (OCC Heightened Standards, FFIEC Handbook, PCI-DSS), industry best practices (AWS Well-Architected Framework Reliability Pillar, NIST SP 800-34), and lessons learned from historical incidents within the organization's operational environment. Deviation from these standards requires formal exception approval through the process defined in Section 16.

An AI-powered validation tool uses this document as its reference corpus to automatically assess SDD compliance. Each requirement is tagged with a unique identifier (e.g., HA-REQ-001) and includes explicit validation criteria that enable automated checking. Human reviewers on the Architecture Review Board use the same criteria for final approval decisions.

1.2 Applicability

This document applies to ALL systems classified as Tier-1 (Business Critical), Tier-2 (Business Important), or Tier-3 (Business Supporting) that operate in production environments. This includes but is not limited to: customer-facing applications, internal business applications, data platforms, integration middleware, batch processing systems, monitoring infrastructure, and shared platform services.

The standards apply regardless of hosting environment: on-premises data center deployments, public cloud (AWS, Azure), hybrid configurations, and managed SaaS platforms where the bank retains architectural control. For SaaS platforms where HA architecture is entirely vendor-managed, the bank MUST obtain documented evidence that the vendor meets equivalent standards through SOC 2 Type II reports, vendor HA SLAs, and contractual uptime guarantees.

Development, testing, and staging environments are EXCLUDED from mandatory compliance but SHOULD mirror production HA patterns to ensure accurate pre-production validation. Performance testing environments MUST replicate the production HA topology to generate representative failover behavior during load testing.

1.3 Compliance Requirements

All new systems entering production MUST demonstrate full compliance with applicable standards (based on their tier classification) prior to receiving Architecture Review Board approval. Compliance is demonstrated through the SDD document which MUST explicitly address each applicable requirement from this standards document.

Existing systems that pre-date this standards document are subject to a compliance remediation timeline. Systems that do not meet current standards MUST submit a remediation plan to the Architecture Review Board within 90 days of the standards' effective date. Remediation MUST be completed within the following timeframes: Tier-1 systems within 6 months, Tier-2 systems within 12 months, and Tier-3 systems within 18 months.

Compliance status is tracked in the Enterprise Architecture Repository and reported monthly to the Technology Risk Committee. Non-compliant systems are flagged as technology risk items and escalated through the enterprise risk management framework. Continued non-compliance beyond the remediation deadline requires CTO-level exception approval with a documented risk acceptance.

1.4 Exceptions Process

In rare cases where full compliance with a specific requirement is technically infeasible or economically disproportionate, a formal exception may be requested. Exceptions MUST be submitted to the Architecture Review Board with the following documentation:

- Specific requirement(s) for which exception is requested (by ID)

Exceptions are granted for a maximum of 12 months and MUST be reviewed at each annual standards review cycle. The Architecture Review Board maintains an exception register that is reported quarterly to the Chief Technology Officer and included in regulatory examination materials when requested by the OCC or other supervisory bodies.

2 System Tiering & Classification

System tiering determines the minimum HA requirements that apply to a given system. Every production system MUST be classified into exactly one tier based on the criteria defined below. Classification is performed during the initial architecture review and MUST be re-evaluated annually or when significant changes occur to the system's business context (e.g., increased customer reach, new regulatory mandate, revenue model change).

System Tiering — HA Requirements by Classification

Requirement	Tier-1 Business Critical	Tier-2 Business Important	Tier-3 Business Supporting
Availability SLA	99.99% (52.6 min/yr)	99.95% (4.38 hr/yr)	99.5% (43.8 hr/yr)
RTO	< 30 seconds	< 5 minutes	< 4 hours
RPO	0 (zero data loss)	< 1 minute	< 1 hour
Deployment	Multi-Region Active-Active	Multi-AZ (min 2, prefer 3)	Single Region N+1 Redundancy
Failover	Automatic (no intervention)	Automatic (validated switchover)	Semi-automatic (operator-initiated)
DR Testing	Monthly (full failover drill)	Quarterly (failover validation)	Semi-annual (recovery test)
Monitoring	24/7 real-time sub-minute alerting	24/7 < 5 min detection	Business hours < 15 min detection
Backup Freq.	Continuous + hourly snapshots	Every 4 hours + daily full	Daily full + weekly archive

Classification Criteria: Revenue impact > $10M/hr = Tier-1 · $1–10M/hr = Tier-2 · < $1M/hr = Tier-3
Customer impact > 1M users = Tier-1 · 100K–1M = Tier-2 · < 100K = Tier-3 · regulatory mandate overrides to Tier-1

Figure 1: System Tiering - HA Requirements by Classification

2.1 Tier-1: Business Critical

Tier-1 systems are those whose unavailability directly and immediately impacts the bank's ability to serve customers, process financial transactions, or meet regulatory obligations. These systems have the highest HA requirements and MUST achieve 99.99% availability (maximum 52.6 minutes of unplanned downtime per year).

[HA-REQ-001]

Tier-1 systems MUST implement an Active-Active or Active-Passive multi-region deployment pattern with automated failover requiring no human intervention for recovery.

[HA-REQ-002]

Tier-1 systems MUST achieve a Recovery Time Objective (RTO) of less than 30 seconds for component-level failures and less than 5 minutes for site-level failures.

[HA-REQ-003]

Tier-1 systems MUST achieve a Recovery Point Objective (RPO) of zero (no data loss) for component-level failures. For site-level failures, RPO MUST NOT exceed 1 minute.

[HA-REQ-004]
Tier-1 systems MUST conduct full failover drills monthly, with results documented and reviewed by the Architecture Review Board.

Examples of Tier-1 systems: Core Banking Platform, Online Banking, Mobile Banking Backend, Payment Processing Gateway, Customer Authentication Service, Real-Time Fraud Detection Engine, Customer Data Platform, ATM/POS Network Gateway.

2.2 Tier-2: Business Important

Tier-2 systems are those whose unavailability causes significant business disruption or customer inconvenience but does not immediately prevent core financial transactions from processing. These systems MUST achieve 99.95% availability (maximum 4.38 hours of unplanned downtime per year).

[HA-REQ-005]
Tier-2 systems MUST implement a multi-Availability Zone deployment pattern with automated failover. Single-AZ deployments are NOT permitted for Tier-2 systems.

[HA-REQ-006]
Tier-2 systems MUST achieve an RTO of less than 5 minutes and an RPO of less than 1 minute for all failure scenarios including AZ-level failures.

[HA-REQ-007]
Tier-2 systems MUST conduct failover validation quarterly, with results documented and any failures remediated within 30 days.

Examples of Tier-2 systems: CRM Platform, Marketing Automation, Document Management, Internal Communications, Customer Onboarding, Credit Decisioning (non-real-time), Regulatory Reporting Platform, Branch Management System.

2.3 Tier-3: Business Supporting

Tier-3 systems support business operations but their temporary unavailability does not materially impact customers or critical business processes. These systems MUST achieve 99.5% availability (maximum 43.8 hours of unplanned downtime per year).

[HA-REQ-008]
Tier-3 systems MUST implement N+1 redundancy at minimum with no single points of failure in the critical path. Automated recovery SHOULD be implemented but operator-initiated failover is acceptable.

[HA-REQ-009]
Tier-3 systems MUST achieve an RTO of less than 4 hours and an RPO of less than 1 hour.

[HA-REQ-010]
Tier-3 systems MUST conduct recovery testing semi-annually, validating that documented procedures achieve the target RTO and RPO.

Examples of Tier-3 systems: Internal Analytics Dashboards, Employee Training Platform, Vendor Management Portal, Asset Management System, Internal Wiki/Knowledge Base, Development Tool Chain (CI/CD), Test Environment Infrastructure.

2.4 Classification Criteria

System classification MUST be determined using the following criteria matrix. A system is classified at the highest tier triggered by ANY single criterion. Regulatory mandate automatically classifies a system as Tier-1 regardless of other criteria.

Criterion	Tier-1 Threshold	Tier-2 Threshold	Tier-3 Threshold
Revenue Impact (per hour)	> $10M / hour	$1M - $10M / hour	< $1M / hour
Customer Impact	> 1 million customers	100K - 1M customers	< 100K customers
Regulatory Mandate	Direct regulatory req.	Indirect regulatory	No regulatory impact
Downstream Dependencies	> 10 Tier-1 consumers	5 - 10 consumers	< 5 consumers
Reputational Risk	National media coverage	Social media coverage	Internal only
Transaction Volume	> 10,000 TPS	1,000 - 10,000 TPS	< 1,000 TPS

3 Active-Active Pattern

3.1 Description and Applicability

The Active-Active pattern deploys multiple identical instances of a system across separate failure domains (Availability Zones or Regions), with all instances actively serving production traffic simultaneously. Traffic is distributed across instances by a load balancer or DNS-based routing, and the failure of any single instance (or group of instances in one failure domain) results in automatic redistribution of traffic to surviving instances with no service interruption.

This pattern is MANDATORY for all Tier-1 systems that process stateless or eventually-consistent workloads. It provides the highest availability (99.99%+) and the fastest recovery (sub-30-second, typically transparent to end users). The pattern requires that application components be designed for horizontal scalability and statelessness -- session state MUST be externalized to a shared state store rather than held in-process.

The Active-Active pattern SHALL be applied at both the regional level (multi-region Active-Active for Tier-1 systems requiring zero RPO) and the AZ level (multi-AZ Active-Active for Tier-2 systems and as the intra-region pattern for Tier-1 systems). The specific topology (2-node, 3-node, N-node) depends on capacity requirements and is determined during the architecture design phase.

3.2 Mandatory Requirements

[HA-REQ-011]
Active-Active deployments MUST have a minimum of 2 active nodes (instances, containers, or pods) in separate failure domains. For Tier-1 systems, a minimum of 3 nodes across 3 failure domains is REQUIRED.

[HA-REQ-012]
All Active-Active nodes MUST sit behind a health-checked load balancer (ALB, NLB, or equivalent). The load balancer MUST perform health checks at intervals not exceeding 10 seconds with an unhealthy threshold of 2 consecutive failures.

[HA-REQ-013]
Application components in an Active-Active configuration MUST be stateless. All session state, user context, and transient data MUST be stored in an external state store (Redis, DynamoDB, or equivalent) that is itself highly available.

[HA-REQ-014]
Active-Active deployments MUST implement graceful degradation such that the failure of N-1 nodes (where N is the total node count) does not result in service failure. The surviving node(s) MUST handle the full production load within acceptable latency targets.

[HA-REQ-015]
Auto-scaling MUST be configured to maintain minimum node counts per failure domain and to respond to load increases within 60 seconds. Scaling policies MUST be pre-tested to confirm they can handle 2x normal peak load.

[HA-REQ-016]
All nodes in an Active-Active cluster MUST run identical software versions. Rolling deployments are permitted, but at no point SHALL more than 1/N of nodes be unavailable during a deployment (where N is the total node count).

3.3 Target SLAs

Metric	Target	Measurement Method
Availability	99.99% (52.6 min/yr max downtime)	Synthetic monitoring from external probes
RTO (component)	< 30 seconds	Time from failure detection to traffic rerouting
RPO	0 (zero data loss)	No in-flight transactions lost during failover
Failover Detection	< 10 seconds	Health check interval x unhealthy threshold
Recovery Validation	< 60 seconds	Time from failover to full health confirmation

3.4 Validation Criteria

An SDD implementing the Active-Active pattern MUST demonstrate compliance with the following validation criteria. These criteria are used by the AI validation tool and the Architecture Review Board to assess SDD conformance:

- SDD MUST explicitly state the number of active nodes and their distribution across failure domains

4 Active-Passive Pattern

4.1 Description and Applicability

The Active-Passive pattern maintains a primary (active) instance that serves all production traffic and one or more secondary (passive/standby) instances that remain synchronized but do not serve traffic under normal conditions. Upon failure of the primary, traffic is redirected to the standby which assumes the active role. This pattern is appropriate for stateful workloads where Active-Active would introduce unacceptable data consistency complexity.

This pattern is MANDATORY for Tier-1 systems that cannot be made stateless (e.g., systems with strong consistency requirements for financial transactions) and for Tier-2 systems where Active-Active is not technically feasible. The Active-Passive pattern provides 99.95% availability with RTOs typically in the 1-5 minute range depending on failover automation and data synchronization latency.

The standby instance MUST be maintained in a warm state (pre-provisioned, configured, and ready to accept traffic within the target RTO). Cold standby (requiring provisioning or significant startup time) is NOT acceptable for Tier-1 or Tier-2 systems. The standby MUST reside in a separate failure domain from the primary (different AZ for intra-region, different region for cross-region Active-Passive).

4.2 Mandatory Requirements

[HA-REQ-017]
The primary and standby MUST reside in separate failure domains. For Tier-1 systems, this MUST be separate AWS regions or separate physical data centers. For Tier-2 systems, separate Availability Zones are the minimum requirement.

[HA-REQ-018]
Data replication between primary and standby MUST be synchronous for Tier-1 systems (RPO = 0) and MAY be asynchronous for Tier-2 systems with documented maximum lag not exceeding 1 minute under normal operating conditions.

[HA-REQ-019]
Failover MUST be automated with no human intervention required for detection and switchover. The failover mechanism MUST include protection against split-brain scenarios (quorum-based consensus, STONITH, or equivalent).

[HA-REQ-020]
The standby MUST be continuously validated through automated health checks that confirm readiness to assume the active role. Standby validation MUST occur at intervals not exceeding 60 seconds.

[HA-REQ-021]
Failback procedures MUST be documented and tested. After a failover event, the original primary MUST be recoverable as a new standby without data loss or extended service disruption. Failback SHOULD be automated but MAY be operator-initiated.

4.3 Target SLAs

Metric	Tier-1 Target	Tier-2 Target

Availability	99.95% (4.38 hr/yr)	99.9% (8.76 hr/yr)
RTO	< 1 minute	< 5 minutes
RPO	0 (synchronous repl.)	< 1 minute (async)
Failover Detection	< 15 seconds	< 30 seconds
Failback Time	< 30 minutes	< 2 hours

4.4 Validation Criteria

An SDD implementing the Active-Passive pattern MUST demonstrate compliance with the following validation criteria:

- SDD MUST identify primary and standby locations with explicit failure domain separation

5 N+1 Redundancy Pattern

5.1 Description and Applicability

The N+1 Redundancy pattern ensures that a system has at least one additional instance or component beyond the minimum required to handle peak production load. If N instances are required to serve peak traffic, then N+1 (or more) instances MUST be deployed. The additional capacity absorbs the loss of any single instance without degradation.

This pattern is the MINIMUM acceptable standard for any production system regardless of tier. Even Tier-3 systems MUST implement N+1 redundancy. For higher tiers, N+1 is a component within the broader Active-Active or Active-Passive pattern. The N+1 pattern alone (without multi-AZ distribution) is only acceptable for Tier-3 systems.

5.2 Capacity Requirements

[HA-REQ-022]
Systems MUST be provisioned such that the loss of any single instance does not cause the remaining instances to exceed 80% of their rated capacity at peak load. This ensures headroom for traffic spikes during degraded operation.

[HA-REQ-023]
Capacity planning MUST account for failure scenarios. The system MUST demonstrate (via load testing) that N-1 instances can sustain peak production load for at least 4 hours without performance degradation exceeding 20% of baseline P99 latency.

[HA-REQ-024]
Auto-scaling MUST be configured to replace failed instances within 5 minutes. Auto-scaling policies MUST be tested quarterly to confirm they function correctly under realistic failure conditions.

The N+1 calculation MUST be performed for each layer of the system independently. If a system has a web tier (3 instances at peak), an application tier (4 instances at peak), and a database tier (2 nodes at peak), then the minimum deployment is: web tier 4 instances, application tier 5 instances, database tier 3 nodes. Each layer MUST independently survive the loss of one component.

5.3 Pre-warming Requirements

[HA-REQ-025]
Standby or auto-scaled instances MUST be pre-warmed: application code loaded, connections established to downstream dependencies, caches populated with essential data, and health checks passing BEFORE the instance is added to the load balancer pool.

[HA-REQ-026]
The pre-warming process MUST complete within 120 seconds of instance launch. Systems that require longer warm-up times MUST maintain additional standby instances to ensure immediate availability during failover.

Pre-warming is critical because a cold instance added to a live load balancer pool can experience cache misses, connection storms, and latency spikes that degrade the overall system. The SDD MUST document the warm-up procedure, expected duration, and the health check criteria that gates traffic routing to the new instance.

5.4 Validation Criteria

- SDD MUST state the number of instances at each tier and demonstrate N+1 at every layer

6 Multi-AZ Deployment Pattern (Cloud)

The Multi-Availability Zone deployment pattern is MANDATORY for all cloud workloads regardless of tier classification. No production system SHALL be deployed to a single Availability Zone. This requirement is non-negotiable and no exceptions will be granted for production systems.

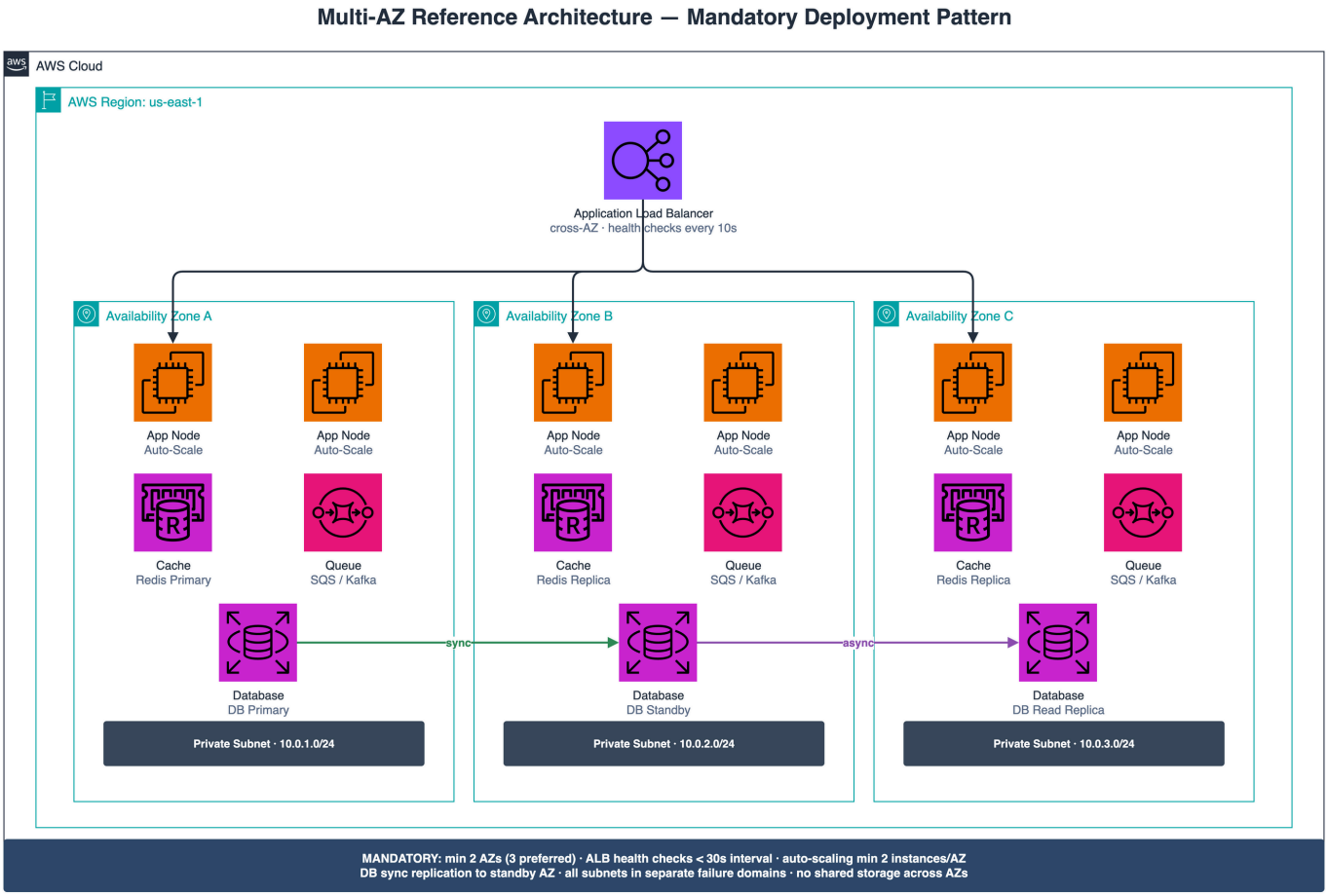


Figure 2: Multi-AZ Reference Architecture - Mandatory Deployment Pattern

6.1 Mandatory Requirements

[HA-REQ-027]

All cloud production workloads MUST be deployed across a minimum of 2 Availability Zones. Tier-1 and Tier-2 systems SHOULD use 3 AZs where the cloud provider supports it.

[HA-REQ-028]

Each Availability Zone MUST host sufficient capacity to serve the full production workload independently in the event that all other AZs become unavailable. For 2-AZ deployments, each AZ MUST handle 100% of load. For 3-AZ deployments, each AZ MUST handle at least 50% of load (allowing for one AZ loss).

[HA-REQ-029]

Instances, containers, or pods MUST be distributed evenly across all AZs using placement groups or anti-affinity rules. No single AZ SHALL host more than ceil(N/AZ_count) + 1 instances of any given service.

[HA-REQ-030]
Deployment automation MUST prevent deployments that would violate AZ distribution requirements. Infrastructure-as-Code templates MUST explicitly specify multi-AZ distribution and MUST fail validation if single-AZ configuration is detected.

6.2 Cross-AZ Load Balancing

[HA-REQ-031]
Application Load Balancers (or equivalent) MUST be configured with cross-zone load balancing enabled. Health checks MUST be configured with a check interval not exceeding 10 seconds and an unhealthy threshold of 2 consecutive failures.

[HA-REQ-032]
Load balancers MUST implement connection draining with a minimum timeout of 30 seconds to allow in-flight requests to complete before removing an unhealthy target from the pool.

Cross-AZ load balancing ensures even distribution of traffic regardless of AZ capacity or health status. The load balancer MUST automatically stop routing traffic to unhealthy AZs (via target health checks) and redistribute to healthy AZs. This behavior MUST be tested during regular failover drills.

6.3 Data Replication Requirements

[HA-REQ-033]
Databases and persistent storage MUST use synchronous replication to at least one additional AZ. For RDS, Multi-AZ with synchronous standby is MANDATORY. For self-managed databases, synchronous replication MUST be configured and monitored.

[HA-REQ-034]
Object storage (S3) MUST use Standard storage class (which provides cross-AZ durability by default). Single-AZ storage classes (S3 One Zone-IA) are PROHIBITED for production data.

[HA-REQ-035]
Caching layers (Redis, Memcached) MUST be deployed with cross-AZ replication. ElastiCache clusters MUST use Multi-AZ with automatic failover enabled.

Data consistency during AZ failover MUST be validated through testing. The SDD MUST document the expected data consistency guarantees during failover (strong consistency, eventual consistency, or potential for brief inconsistency window) and any application-level compensating logic.

6.4 Subnet Design Standards

[HA-REQ-036]
Each AZ MUST have dedicated subnets for each functional tier: public (load balancers, NAT gateways), private application (compute instances), and private data (databases, caches). Subnets MUST NOT span multiple AZs.

[HA-REQ-037]
Subnet CIDR allocation MUST provide sufficient IP address space for auto-scaling events. Each subnet MUST have a minimum of 2x the expected maximum instance count in available IP addresses to accommodate scaling without address exhaustion.

The subnet design MUST be documented in the SDD with a network diagram showing the VPC layout, subnet CIDRs, routing tables, NAT gateway placement, and security group boundaries. The diagram MUST clearly show the separation of failure domains.

7 Multi-Region Deployment Pattern (Cloud)

7.1 When Required

Multi-region deployment is MANDATORY for Tier-1 systems. It is REQUIRED when any of the following conditions apply: the system is classified as Tier-1 (Business Critical), regulatory requirements mandate geographic redundancy, the system processes international traffic requiring proximity, or the business impact of a full regional outage exceeds acceptable thresholds (> $10M per hour).

[HA-REQ-038]
Tier-1 systems MUST be deployed across a minimum of 2 geographically separate regions with independent failure domains. The secondary region MUST be capable of assuming full production load within the documented RTO.

[HA-REQ-039]
The primary and secondary regions MUST be geographically separated by at least 200 miles to protect against regional disasters (earthquakes, hurricanes, widespread power outages). For Example Organization, the standard region pair is us-east-1 (primary) and us-west-2 (secondary).

7.2 Primary/Secondary Configuration

[HA-REQ-040]
The secondary region MUST maintain a continuously running deployment (warm standby at minimum) with infrastructure pre-provisioned and data replication active. Pilot light configurations (infrastructure provisioned on-demand) are NOT acceptable for Tier-1 systems.

[HA-REQ-041]
The secondary region MUST run continuous synthetic health checks that validate end-to-end system functionality. These checks MUST exercise the full application stack including database reads, external integrations, and authentication flows.

[HA-REQ-042]
Infrastructure in both regions MUST be deployed from the same Infrastructure-as-Code source (Terraform, CloudFormation, or equivalent). Configuration drift between regions MUST be detected automatically and remediated within 24 hours.

7.3 DNS Health Check Requirements

[HA-REQ-043]
DNS-based failover (Route 53 health checks or equivalent) MUST be configured with health check intervals not exceeding 10 seconds and a failover threshold of 3 consecutive failures (triggering failover within 30 seconds of failure detection).

[HA-REQ-044]
DNS TTL for failover records MUST NOT exceed 60 seconds. Lower TTLs (10-30 seconds) are RECOMMENDED for Tier-1 systems to minimize the window during which clients may route to the failed region.

[HA-REQ-045]
Health check endpoints MUST validate full application stack health (not just TCP connectivity). The health check MUST

return unhealthy if any critical dependency (database, cache, external service) is unavailable.

7.4 IaC Parity Requirements

[HA-REQ-046]

All infrastructure MUST be defined in version-controlled Infrastructure-as-Code. Manual console changes are PROHIBITED in production regions. Detected drift MUST trigger a P2 alert and be remediated within 24 hours.

[HA-REQ-047]

DR drills MUST be conducted monthly for Tier-1 systems. Each drill MUST include a full regional failover, validation of all system functions in the secondary region, and a controlled failback to the primary. Drill results MUST be documented with any issues remediated within 14 days.

The SDD MUST include a DR runbook that documents the step-by-step failover procedure, including automated steps, manual verification checkpoints, communication protocols, and rollback procedures. The runbook MUST be validated during each DR drill.

8 Database HA Standards

Databases are the most critical component in the HA architecture because they hold persistent state that cannot be easily reconstructed. Database HA failures result in data loss, which is the most severe category of outage. The following standards apply to all production databases regardless of engine (relational, document, key-value, graph).

8.1 Replication Requirements

[HA-REQ-048]
Tier-1 database systems MUST implement synchronous replication to at least one standby in a separate failure domain. The synchronous standby MUST acknowledge every transaction before the primary confirms the commit to the application.

[HA-REQ-049]
Tier-2 database systems MUST implement replication (synchronous or asynchronous) to at least one standby. If asynchronous, the maximum replication lag under normal operating conditions MUST NOT exceed 1 second (measured as P99 over a 24-hour period).

[HA-REQ-050]
Replication lag MUST be continuously monitored with alerting configured at the following thresholds: Warning at 5 seconds lag, Critical at 30 seconds lag. Lag exceeding 5 minutes MUST trigger an automated incident.

[HA-REQ-051]
For multi-region deployments, cross-region replication MUST be configured with documented RPO targets. Cross-region replication lag MUST be monitored and reported separately from intra-region replication.

8.2 Automated Failover

[HA-REQ-052]
Database failover MUST be automated with no human intervention required for detection and switchover. For cloud-managed databases (RDS, Aurora), native Multi-AZ failover MUST be enabled. For self-managed databases, an automated failover orchestrator (Patroni, Orchestrator, or equivalent) MUST be deployed.

[HA-REQ-053]
The failover mechanism MUST include split-brain prevention. For consensus-based systems, a minimum of 3 nodes with quorum-based leader election is REQUIRED. For two-node configurations, a witness/arbiter node MUST be deployed in a third failure domain.

[HA-REQ-054]
Database failover MUST complete within 30 seconds for Tier-1 systems and within 60 seconds for Tier-2 systems. This includes failure detection, leader election, and DNS/endpoint propagation to application connection pools.

8.3 Backup & PITR

[HA-REQ-055]
All production databases MUST implement continuous backup with Point-in-Time Recovery (PITR) capability. PITR granularity MUST be 5 minutes or better for Tier-1 systems and 1 hour or better for Tier-2 systems.

[HA-REQ-056]

Backup retention MUST meet the following minimums: Tier-1: 35 days of PITR + 7 years of monthly snapshots. Tier-2: 14 days of PITR + 3 years of monthly snapshots. Tier-3: 7 days of daily snapshots + 1 year of monthly snapshots.

[HA-REQ-057]
Backup restoration MUST be tested quarterly for Tier-1 systems and semi-annually for Tier-2/3 systems. Tests MUST validate both full restoration and point-in-time recovery to a specific timestamp. Test results MUST be documented with measured restoration time.

[HA-REQ-058]
Backups MUST be stored in a separate failure domain from the primary database. For cloud databases, backups MUST be replicated to a different region. Backup encryption MUST use customer-managed keys with key rotation every 90 days.

8.4 Connection Pooling Standards

[HA-REQ-059]
Applications MUST use connection pooling for all database access. Connection pools MUST be configured with maximum connection limits that prevent database exhaustion under load (maximum connections per pool MUST NOT exceed 80% of database max_connections divided by the number of application instances).

[HA-REQ-060]
Connection pools MUST implement connection validation (test-on-borrow or equivalent) to detect and replace stale connections after failover. The validation interval MUST NOT exceed 30 seconds. Pools MUST support automatic reconnection to the new primary after a failover event without application restart.

Read replicas SHOULD be used to offload read-heavy workloads from the primary database. When read replicas are used, the application MUST implement read-after-write consistency where required (directing reads that follow a recent write to the primary rather than a potentially-lagging replica).

9 Queue-Based Resilience Pattern

Message queues and event streaming platforms provide temporal decoupling between producers and consumers, enabling resilience through buffering. When a downstream consumer is temporarily unavailable, messages are retained in the queue until the consumer recovers. This pattern is MANDATORY for all asynchronous inter-service communication in Tier-1 and Tier-2 systems.

9.1 Dead Letter Queue Requirements

[HA-REQ-061]
Every queue/topic MUST have an associated Dead Letter Queue (DLQ) configured. Messages that fail processing after the maximum retry count MUST be routed to the DLQ rather than being silently dropped.

[HA-REQ-062]
The maximum retry count before DLQ routing MUST be configured between 3 and 5 attempts with exponential backoff. The initial retry delay MUST be at least 1 second, with a maximum backoff of 5 minutes.

[HA-REQ-063]
DLQ monitoring MUST be configured with alerts triggering when DLQ depth exceeds defined thresholds: Warning at 100 messages, Critical at 1000 messages. DLQ messages MUST be reviewed and resolved within 24 hours of arrival.

[HA-REQ-064]
A DLQ replay mechanism MUST be implemented to allow reprocessing of DLQ messages after the root cause is resolved. The replay mechanism MUST support selective replay (by message attribute or time range) without requiring full queue replay.

9.2 Message Retention

[HA-REQ-065]
Message retention MUST be configured to survive extended consumer outages. Minimum retention periods: Tier-1 queues: 14 days, Tier-2 queues: 7 days, Tier-3 queues: 3 days. For Kafka topics, retention MUST be at least 7 days for all tiers.

[HA-REQ-066]
Queue/topic storage MUST be replicated across multiple failure domains (AZs). For SQS, standard replication is acceptable. For Kafka/MSK, a replication factor of 3 across 3 AZs is MANDATORY. For self-managed queues, synchronous replication to at least 2 nodes is REQUIRED.

9.3 Consumer Idempotency

[HA-REQ-067]
All message consumers MUST be idempotent -- processing the same message multiple times MUST produce the same result as processing it once. This is REQUIRED because at-least-once delivery is the default guarantee for distributed message systems.

[HA-REQ-068]
Idempotency MUST be implemented using one of the following approaches: unique message ID deduplication (with a deduplication window at least 2x the retention period), database-level upsert operations, or application-level version

checking.

The SDD MUST document the idempotency strategy for each consumer, the deduplication mechanism, and how duplicate processing is detected and prevented. The strategy MUST be validated through chaos testing that deliberately re-delivers messages.

10 Circuit Breaker Pattern

The Circuit Breaker pattern prevents cascading failures by detecting when a downstream dependency is failing and temporarily stopping calls to that dependency. This allows the failing service to recover without being overwhelmed by retry storms, and allows the calling service to fail fast rather than accumulating timeouts. This pattern is MANDATORY for all service-to-service communication in Tier-1 and Tier-2 systems.

10.1 Configuration Standards

[HA-REQ-069]
Circuit breakers MUST be implemented on all outbound service calls to external dependencies including: downstream microservices, databases, caches, third-party APIs, and shared platform services. No service-to-service call SHALL be made without circuit breaker protection.

[HA-REQ-070]
Circuit breaker thresholds MUST be configured as follows: Failure threshold to OPEN the circuit: 50% failure rate over a 10-second sliding window with a minimum of 20 requests. Half-open retry: 1 request every 30 seconds. Recovery threshold to CLOSE: 5 consecutive successful requests in half-open state.

[HA-REQ-071]
Timeout values for downstream calls MUST be explicitly configured (no infinite timeouts permitted). Timeout MUST be set at the P99 latency of the downstream service plus a buffer of 20%. If the downstream service P99 is unknown, a maximum timeout of 5 seconds MUST be used until baseline metrics are established.

[HA-REQ-072]
Circuit breaker state transitions (CLOSED -> OPEN, OPEN -> HALF-OPEN, HALF-OPEN -> CLOSED) MUST be logged and emit metrics. An alert MUST fire when any circuit breaker transitions to OPEN state, indicating a downstream failure in progress.

10.2 Fallback Behavior

[HA-REQ-073]
Every circuit breaker MUST have a defined fallback behavior that is executed when the circuit is OPEN. Acceptable fallback behaviors include: returning cached data (with staleness indicator), returning a degraded response (subset of data), queuing the request for later processing, or returning a meaningful error to the caller.

[HA-REQ-074]
Fallback behavior MUST NOT include retrying the same failed dependency (which would defeat the purpose of the circuit breaker). If retry is needed, it MUST target an alternative endpoint or use a different communication path.

The SDD MUST document the fallback behavior for each circuit breaker, including the user-facing impact of operating in fallback mode and the maximum duration the system can operate with the circuit open before manual intervention is required.

10.3 Monitoring Requirements

[HA-REQ-075]
Circuit breaker metrics MUST be reported to the monitoring system including: circuit state (open/closed/half-open),

failure count, success count, fallback invocation count, and time spent in open state. Dashboards MUST visualize circuit health for all downstream dependencies.

Circuit breaker configurations (thresholds, timeouts, fallback behaviors) MUST be externalized to configuration (not hardcoded) to allow tuning without code deployments. Configuration changes MUST be auditable and reversible.

11 Network Resilience Standards

Network connectivity is a foundational dependency for all distributed systems. Network failures can manifest as total loss of connectivity, increased latency, packet loss, or asymmetric routing. The following standards ensure network resilience across all production environments.

11.1 Redundant Connectivity

[HA-REQ-076]
All production network paths MUST have at least two independent physical paths with no shared single point of failure (no shared router, switch, fiber path, or provider). For on-premises to cloud connectivity (Direct Connect, ExpressRoute), dual circuits from different providers terminating at different facilities are MANDATORY.

[HA-REQ-077]
Cross-data-center links MUST be provisioned from at least two telecommunications providers using diverse physical paths. Path diversity MUST be verified through provider documentation showing separate fiber routes, and validated annually.

[HA-REQ-078]
Network failover MUST be automated via BGP with Bidirectional Forwarding Detection (BFD) for sub-second failure detection. Failover between redundant paths MUST complete within 3 seconds of failure detection.

11.2 DNS Redundancy

[HA-REQ-079]
DNS resolution MUST use redundant DNS providers or multi-AZ DNS infrastructure. Systems MUST NOT depend on a single DNS resolver. Client-side DNS caching with reasonable TTLs (60-300 seconds) MUST be implemented to survive brief DNS outages.

[HA-REQ-080]
DNS records for production services MUST be hosted on managed DNS services with SLA guarantees of 100% availability (Route 53, Cloud DNS, or equivalent). Self-hosted DNS servers are PROHIBITED for production service resolution.

11.3 Load Balancer HA

[HA-REQ-081]
Load balancers MUST be deployed in a highly available configuration spanning multiple AZs. Single-instance load balancers are PROHIBITED. For cloud environments, managed load balancers (ALB, NLB) with native multi-AZ support are REQUIRED.

[HA-REQ-082]
For on-premises load balancers (F5, Citrix, HAProxy), an active-passive pair at minimum is REQUIRED with stateful failover (session table synchronization). Health check intervals MUST NOT exceed 5 seconds with automatic failover within 10 seconds.

[HA-REQ-083]
Load balancer capacity MUST be pre-provisioned to handle peak traffic without scaling delays. For AWS ALB/NLB,

pre-warming requests MUST be submitted to AWS for any anticipated traffic increase exceeding 50% above baseline.

12 Monitoring & Alerting Standards

Effective monitoring and alerting is essential for achieving HA targets. A system cannot recover from a failure it does not detect. The following standards define the minimum monitoring requirements that every production system MUST implement.

12.1 Health Check Requirements

[HA-REQ-084]
Every production service MUST implement a dedicated health check endpoint that validates the service's ability to perform its primary function. The health check MUST verify connectivity to all critical dependencies (database, cache, downstream services) and return unhealthy if any critical dependency is unavailable.

[HA-REQ-085]
Health checks MUST distinguish between liveness (process is running) and readiness (process is ready to accept traffic). A service that is live but not ready MUST NOT receive production traffic. Load balancers MUST use readiness checks for routing decisions.

[HA-REQ-086]
Health check frequency MUST meet the following minimums: Load balancer health checks: every 10 seconds. Synthetic external monitoring: every 30 seconds. Internal component health checks: every 60 seconds. Dependency health validation: every 30 seconds.

[HA-REQ-087]
Health checks MUST have timeout values configured. A health check that does not respond within 5 seconds MUST be treated as a failure. Health check endpoints MUST be lightweight and MUST NOT perform expensive operations that could contribute to system overload.

12.2 Alert Threshold Standards

[HA-REQ-088]
Alerting MUST be configured at three severity levels: Critical (P1: immediate response required, 24/7 paging), High (P2: response within 15 minutes during business hours, 30 minutes off-hours), and Warning (P3: response within 4 business hours).

The following standard alert thresholds MUST be implemented for all production systems:

Metric	Warning (P3)	High (P2)	Critical (P1)
Error Rate	> 1%	> 5%	> 10%
P99 Latency	> 2x baseline	> 5x baseline	> 10x baseline
CPU Utilization	> 70%	> 85%	> 95%
Memory Utilization	> 75%	> 85%	> 95%
Disk Utilization	> 70%	> 85%	> 95%
Replication Lag	> 5 seconds	> 30 seconds	> 5 minutes
Queue Depth	> 10x normal	> 50x normal	> 100x normal

Health Check Failures	> 1 consecutive	> 3 consecutive	> 5 consecutive
Connection Pool Usage	> 70%	> 85%	> 95%

[HA-REQ-089]
Alerts MUST include escalation procedures: P1 alerts MUST escalate to the on-call engineer within 5 minutes if not acknowledged, to the team lead within 15 minutes, and to the engineering director within 30 minutes. Escalation MUST be automated (PagerDuty or equivalent).

12.3 Observability Requirements

[HA-REQ-090]
All production systems MUST implement the three pillars of observability: metrics (quantitative measurements), logs (event records), and distributed traces (request flow across services). These MUST be correlated via shared identifiers (trace ID, request ID).

[HA-REQ-091]
Metrics MUST be retained for a minimum of 13 months (to enable year-over-year comparison). Logs MUST be retained for a minimum of 90 days in hot storage (searchable) and 1 year in cold storage (archive). Traces MUST be sampled at a minimum 1% rate and retained for 30 days.

[HA-REQ-092]
Dashboards MUST be maintained for each production system showing: real-time health status, error rates, latency distributions, throughput, saturation metrics, and dependency health. Dashboards MUST be accessible to all on-call engineers without additional authentication steps.

13 Backup & Recovery Standards

Backup and recovery capabilities provide the last line of defense against data loss. While HA mechanisms protect against hardware failures and brief outages, backups protect against data corruption, accidental deletion, ransomware, and catastrophic site failures. The following standards are MANDATORY for all production data stores.

13.1 Backup Frequency by Tier

Tier	PITR / Continuous	Snapshot Frequency	Full Backup	Off-site Copy
Tier-1	MANDATORY (continuous	Every 1 hour	Daily	Cross-region, real-time
Tier-2	REQUIRED (max 5-min gap)	Every 4 hours	Daily	Cross-region, daily
Tier-3	RECOMMENDED	Every 12 hours	Daily	Same region, different AZ

[HA-REQ-093]

Tier-1 systems MUST implement continuous backup (WAL archiving, redo log shipping, or equivalent) enabling point-in-time recovery to any second within the retention window. Additionally, hourly snapshots MUST be taken for rapid bulk recovery.

[HA-REQ-094]

All backups MUST be encrypted using customer-managed encryption keys. Backup encryption keys MUST be stored separately from the backup data (different AWS account or separate key management infrastructure). Key loss MUST NOT result in unrecoverable backups (key escrow procedures MUST be documented).

[HA-REQ-095]

Backup integrity MUST be validated automatically. Each backup MUST include a checksum that is verified during the backup process. Monthly integrity validation MUST perform a trial restoration of a random backup to confirm recoverability.

13.2 Retention Requirements

[HA-REQ-096]

Backup retention MUST comply with both operational and regulatory requirements. The minimum operational retention is: PITR/continuous: 35 days (Tier-1), 14 days (Tier-2), 7 days (Tier-3). Snapshot retention: 90 days for all tiers. Regulatory retention: as mandated by applicable regulations (minimum 7 years for financial records).

[HA-REQ-097]

Backup deletion MUST be protected against accidental or malicious removal. Deletion locks (S3 Object Lock, immutable storage, or equivalent) MUST be applied to backups within the regulatory retention period. Backup deletion MUST require multi-party authorization (at least 2 approvers).

13.3 Recovery Testing

[HA-REQ-098]

Recovery testing MUST be conducted at the following frequencies: Tier-1: monthly (full restoration to separate environment, validated against production data). Tier-2: quarterly. Tier-3: semi-annually. Recovery tests MUST validate the documented RTO and RPO targets are achievable.

[HA-REQ-099]

Recovery test results MUST document: actual recovery time (compared to RTO target), actual data loss (compared to RPO target), any issues encountered, and remediation actions for any failures. Results MUST be retained for 3 years and available for regulatory examination.

[HA-REQ-100]

Recovery procedures MUST be documented in runbooks that are tested during each recovery drill. Runbooks MUST include step-by-step procedures, expected timelines for each step, verification checkpoints, escalation contacts, and rollback procedures if recovery fails.

14 Security & Encryption Standards (HA-Related)

Security controls MUST be designed to support high availability rather than impede it. The following standards address the intersection of security and availability, ensuring that encryption, certificate management, and access control do not create single points of failure or extend recovery times.

14.1 Key Availability

[HA-REQ-101]
Encryption keys MUST be available in all failure domains where the encrypted data resides. For multi-AZ deployments, KMS keys MUST be regional (available across all AZs). For multi-region deployments, KMS multi-region keys MUST be used to ensure decryption capability in the DR region.

[HA-REQ-102]
Key management systems MUST provide 99.999% availability (five nines). AWS KMS satisfies this requirement natively. For on-premises HSM deployments, HA clustering with at least 3 HSM nodes across 2 failure domains is REQUIRED.

[HA-REQ-103]
Key rotation MUST NOT cause service disruption. Applications MUST support decryption with the previous key version for at least 24 hours after rotation to allow propagation. Key rotation MUST be automated and occur at least every 90 days.

14.2 Certificate Management

[HA-REQ-104]
TLS certificates MUST be managed through automated certificate management (ACM, Let's Encrypt, or equivalent) that handles renewal without manual intervention. Certificate expiry MUST trigger alerts at 30 days, 14 days, and 7 days before expiration.

[HA-REQ-105]
Certificate renewal MUST be tested in non-production environments before production rotation. Certificates MUST be deployed to all instances within a service simultaneously to prevent mixed-certificate states that can cause TLS handshake failures.

[HA-REQ-106]
Mutual TLS (mTLS) configurations MUST include certificate pinning fallback mechanisms to prevent service disruption if a certificate is revoked. Certificate revocation checking MUST use OCSP stapling rather than synchronous CRL checks to avoid availability dependencies on the certificate authority.

14.3 Access Control During DR

[HA-REQ-107]
Access control mechanisms MUST function correctly in DR scenarios. Authentication services (LDAP, AD, OIDC providers) MUST be available in the DR region. If the primary identity provider is unavailable, a pre-configured fallback authentication mechanism MUST allow authorized personnel to access DR systems.

[HA-REQ-108]

Break-glass procedures MUST be documented for scenarios where normal access control is unavailable during a DR event. Break-glass credentials MUST be stored in a secure, offline medium (sealed envelope in physical safe, or equivalent) and tested semi-annually.

The SDD MUST document how access control is maintained during failover scenarios, including any temporary elevated privileges required for DR operations and the process for revoking them after recovery.

15 Validation Checklist

The following checklist is used by both the AI validation tool and human reviewers to assess SDD compliance with these standards. Each item is categorized by severity (Critical, High, Medium) and indicates whether it can be validated automatically or requires manual review.

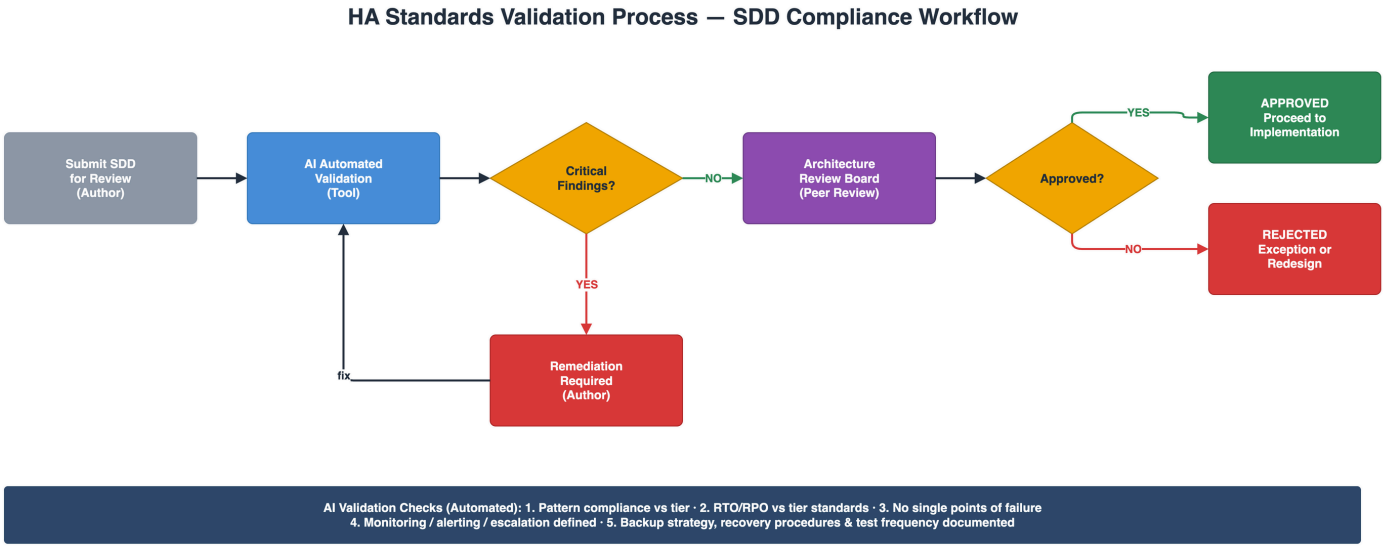


Figure 3: HA Standards Validation Process - SDD Compliance Workflow

15.1 Critical Items (Must Pass - Blocks Approval)

Failure of any Critical item results in automatic rejection of the SDD. The system MUST NOT proceed to production deployment until all Critical items are resolved.

ID	Check	Validation Method
C-01	System tier explicitly stated and classification criteria documented	Automated - text search
C-02	HA pattern matches tier requirement (Active-Active for T1, Multi-AZ for T2)	Automated - pattern match
C-03	RTO and RPO targets stated and meet tier minimums	Automated - value comparison
C-04	No single points of failure identified in architecture diagram	Manual - architecture review
C-05	Multi-AZ deployment confirmed (min 2 AZs for all cloud workloads)	Automated - infra config check
C-06	Database replication configured with appropriate mode (sync/async)	Automated - config check

C-07	Automated failover mechanism documented with split-brain prevention	Manual - design review
C-08	Backup strategy documented with frequency meeting tier requirements	Automated - schedule check
C-09	Health check endpoints defined for all services	Automated - endpoint check
C-10	DR testing frequency meets tier requirement	Automated - schedule check

15.2 High Priority Items (Should Pass - Requires Justification)

Failure of High priority items requires documented justification and a remediation timeline. The ARB may grant conditional approval with a mandatory remediation plan.

ID	Check	Validation Method
H-01	Circuit breakers configured on all outbound service calls	Automated - code/config scan
H-02	Dead Letter Queues configured for all message consumers	Automated - infra config
H-03	Monitoring covers all three pillars (metrics, logs, traces)	Manual - observability review
H-04	Alert thresholds defined at Warning, High, and Critical levels	Automated - config check
H-05	Connection pooling configured with appropriate limits	Automated - config check
H-06	Auto-scaling configured with tested scaling policies	Automated - infra config
H-07	Network redundancy documented (dual paths, diverse providers)	Manual - network review
H-08	Encryption keys available in all failure domains	Automated - KMS config check
H-09	Certificate management automated with expiry alerting	Automated - cert check
H-10	Load testing validates N-1 capacity handling	Manual - test results review

15.3 Medium Priority Items (Recommended - Best Practice)

Medium priority items represent best practices. Failure does not block approval but is noted for future improvement and tracked in the technical debt register.

ID	Check	Validation Method
M-01	Pre-warming procedures documented for auto-scaled instances	Manual - doc review
M-02	Chaos testing conducted and results documented	Manual - test results
M-03	Runbooks exist for all failure scenarios with tested procedures	Manual - doc review
M-04	Consumer idempotency strategy documented for all queue consumers	Manual - design review
M-05	Read replica strategy documented for read-heavy workloads	Manual - design review
M-06	Capacity planning shows 18-month forward projection	Manual - capacity review
M-07	Graceful degradation behavior documented for all dependencies	Manual - design review
M-08	Cross-region IaC parity verified with drift detection	Automated - IaC check

M-09	Backup restoration tested within last quarter	Automated - test log check
M-10	SLI/SLO definitions align with published SLA targets	Manual - alignment review

16 Compliance & Governance

16.1 Architecture Review Board

The Architecture Review Board (ARB) is the governing body responsible for approving all production system deployments and ensuring compliance with these standards. No system SHALL proceed to production deployment without ARB approval. The ARB meets weekly and reviews all submitted SDDs using the validation checklist defined in Section 15.

The ARB review process follows these steps: (1) Author submits completed SDD with self-assessment against validation checklist. (2) AI validation tool performs automated checks and generates a compliance report. (3) ARB members review the SDD, compliance report, and any flagged items. (4) ARB issues one of three decisions: Approved (proceed to implementation), Conditionally Approved (proceed with documented remediation plan), or Rejected (return to author for redesign).

[HA-REQ-109]
All SDDs MUST undergo AI-assisted validation before ARB submission. The AI validation report MUST be included with the SDD submission. Authors MUST address all Critical findings before submission.

16.2 Exception Process

Exceptions to these standards follow the process defined in Section 1.4. Additionally, the following governance rules apply to exceptions:

[HA-REQ-110]
Exceptions MUST be time-bounded (maximum 12 months) with a committed remediation date. Perpetual exceptions are NOT permitted. If remediation is not achieved by the committed date, the exception MUST be re-submitted with updated justification.

[HA-REQ-111]
The exception register MUST be reviewed quarterly by the Technology Risk Committee. Aggregate exception data (count by standard, count by tier, aging) MUST be reported to the CTO monthly.

[HA-REQ-112]
Exceptions for Tier-1 systems require approval from the CTO or designated delegate. Exceptions for Tier-2 systems require VP-level approval. Exceptions for Tier-3 systems require Director-level approval.

16.3 Regulatory Alignment

These standards are designed to satisfy or exceed the following regulatory requirements. Changes to regulatory guidance MUST trigger a review of affected standards within 90 days of the guidance effective date.

Regulation	Relevant Requirements	Standards Alignment
OCC Heightened Standards (12 CFR 30 App D)	IT risk management, operational resilience	Sections 2-7 (tiering, HA patterns, DR)
FFIEC IT Examination Handbook	Business continuity, IT operations	Sections 5-8 (redundancy, database, backup)

PCI-DSS v4.0	Requirement 12.10 (incident response)	Sections 11-13 (monitoring, alerting, recovery)
NIST SP 800-34 Rev. 1	Contingency planning	Sections 6-7 (multi-AZ, multi-region)
OCC 2023-17 (Third-Party Risk)	Third-party resilience requirements	Section 10 (circuit breaker), Section 11 (network)
DORA (EU - cross-border)	Digital operational resilience	Full document alignment

16.4 Annual Review Requirements

[HA-REQ-113]

This standards document MUST be reviewed annually by the Enterprise Architecture team and updated to reflect changes in technology landscape, regulatory requirements, and lessons learned from incidents. The annual review MUST be completed within 60 days of the review trigger date.

[HA-REQ-114]

All production systems MUST undergo an annual HA posture assessment against the current version of these standards. Systems that no longer comply due to standards updates MUST submit a remediation plan within 90 days.

The annual review process considers: (1) Incidents from the past year and their root causes. (2) New regulatory guidance or examination findings. (3) Technology changes (new cloud services, deprecated patterns). (4) Industry best practice evolution. (5) Feedback from ARB reviews and exception patterns. (6) Results from DR drills and recovery tests across all tiers.

Appendix A: Reference Diagrams

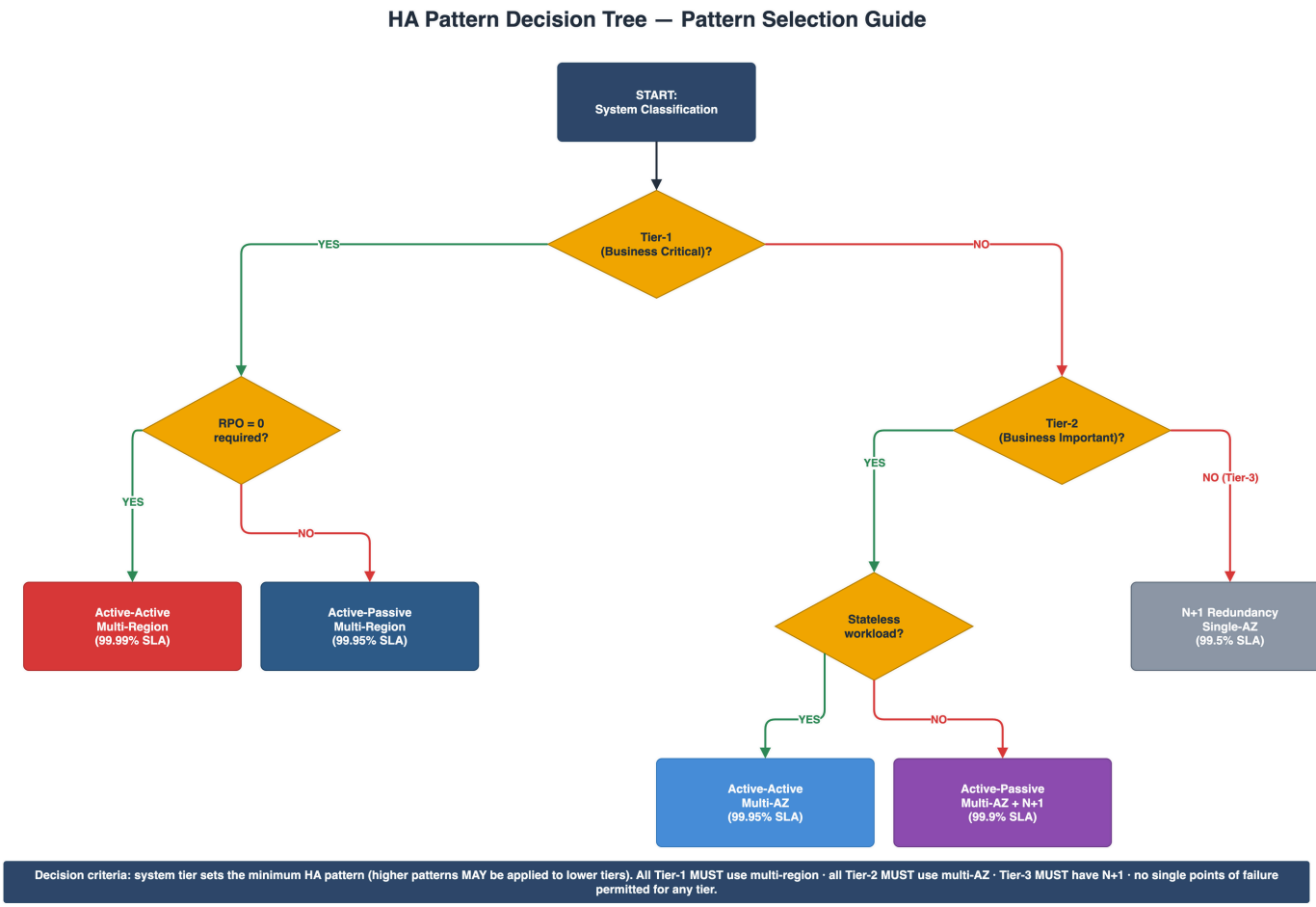


Figure A-1: HA Pattern Decision Tree - Pattern Selection Guide

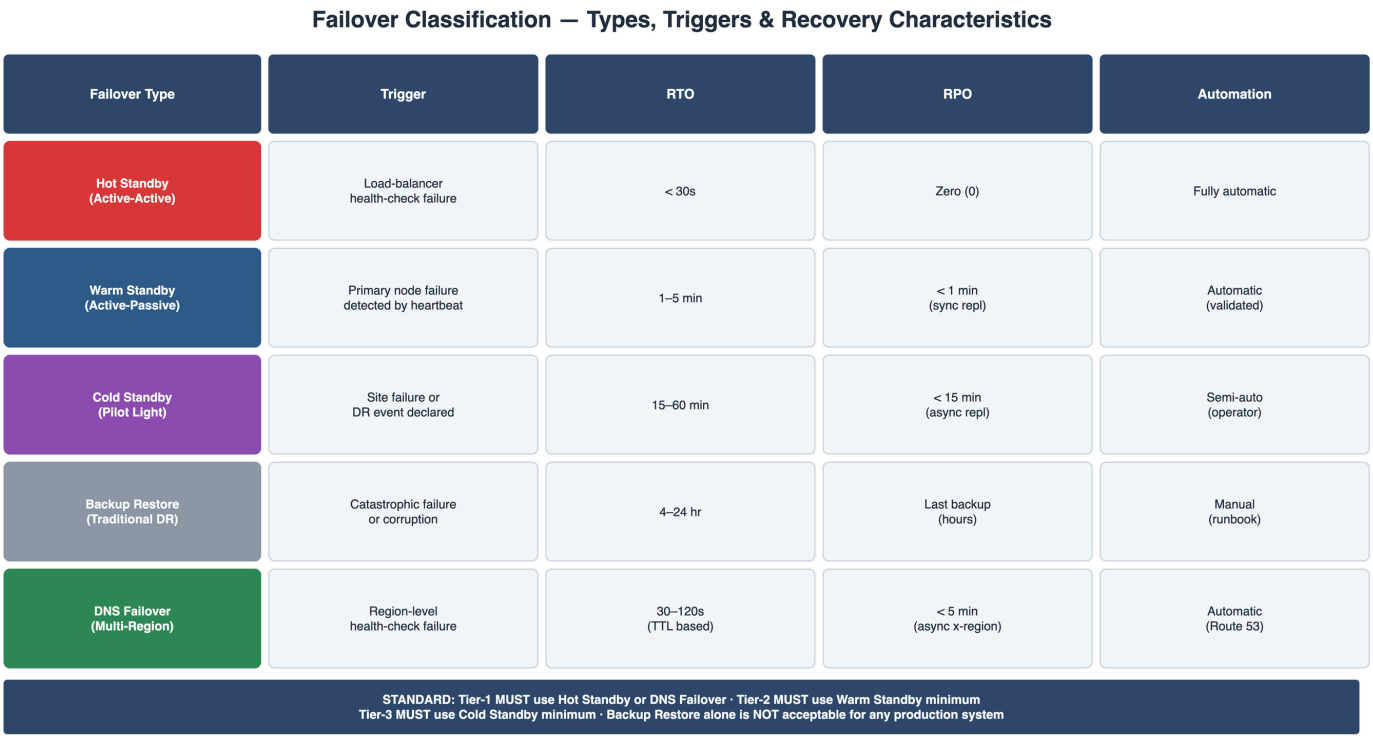


Figure A-2: Failover Classification - Types, Triggers & Recovery Characteristics

Appendix B: References & Source Documents

- AWS Well-Architected Framework - Reliability Pillar (2024 Edition)

Appendix C: Glossary of Terms

Term	Definition
AZ	Availability Zone - isolated location within a cloud region
BFD	Bidirectional Forwarding Detection - sub-second failure detection protocol
BGP	Border Gateway Protocol - routing protocol for network failover
DLQ	Dead Letter Queue - queue for messages that failed processing
DR	Disaster Recovery - restoration of services after a major failure
HA	High Availability - ability to remain operational during component failures
IaC	Infrastructure as Code - version-controlled infrastructure definitions
MTBF	Mean Time Between Failures - average time between system failures
MTTR	Mean Time To Recovery - average time to restore service after failure
PITR	Point-in-Time Recovery - ability to restore data to a specific moment
RPO	Recovery Point Objective - maximum acceptable data loss in time
RTO	Recovery Time Objective - maximum acceptable service restoration time
SLA	Service Level Agreement - contractual availability commitment
SLI	Service Level Indicator - quantitative measure of service performance
SLO	Service Level Objective - target value for an SLI
SPOF	Single Point of Failure - component whose failure causes system failure
WAL	Write-Ahead Log - database durability mechanism for PITR

Document Approval

This document has been reviewed and approved by the following stakeholders. Their signatures indicate agreement that the standards defined herein represent the minimum acceptable HA posture for Example Organization's production systems.

Role	Name	Date	Status
Chief Technology Officer	D. Williams	04/28/2026	Approved
VP Enterprise Architecture	J. Morrison	04/25/2026	Approved
VP Infrastructure Engineering	R. Torres	04/22/2026	Approved
Director Cloud Architecture	A. Kapoor	04/20/2026	Approved
Director Security Architecture	M. Singh	04/21/2026	Approved
Head of Technology Risk	S. Anderson	04/23/2026	Approved
Chief Information Security Officer	P. Rodriguez	04/24/2026	Approved

This document becomes effective upon CTO approval and supersedes all previous versions of the HA Reference Patterns & Standards document. Questions regarding interpretation or application of these standards should be directed to the Enterprise Architecture team at architecture@example.com.